A Benchmarked Pipeline for Ferroptosis-Associated Prognosis, Classification and Drug-Response Analysis in Colon Cancer: Evidence Against Gene-Set Specificity

Mayank Bhardwaj*
Department of Computer Science and Engineering
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology, India

Sovan Sekar Sharma
Department of Computer Science and Engineering
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology, India

Bhuvnesh KV
Department of Computer Science and Engineering
School of Computer Science and Engineering (SCOPE)
Vellore Institute of Technology, India

Abstract

Ferroptosis, an iron-dependent form of regulated cell death, is widely proposed as a source of prognostic and predictive biomarkers in colorectal cancer. Reported ferroptosis signatures are rarely compared with equally sized random gene sets, so apparent performance may reflect generic transcriptomic signal rather than ferroptosis biology. This study presents a fully scripted pipeline that applies one design rule to every analysis: each result obtained with a 36-gene ferroptosis panel is compared with 200 random gene sets of the same size that pass through the identical procedure. The pipeline covers (i) a ridge-penalised Cox prognosis model trained on TCGA-COAD (n = 434 with valid overall survival) and frozen before evaluation in three independent GEO cohorts; (ii) classification of mismatch-repair deficiency (dMMR); (iii) unsupervised ferroptosis subtypes; and (iv) prediction of cell-line GPX4 dependence and chemotherapy IC_{50} from expression. In external validation the frozen model gave concordance indices of 0.539 (GSE39582), 0.486 (GSE17536) and 0.581 (GSE14333), none distinguishable from random gene sets (empirical p = 0.338, 0.831, 0.343), and below a clinical model of stage and age (0.675 and 0.747 in the two OS cohorts). The frozen score was not significantly associated with outcome in any single cohort (pooled OS hazard ratio per SD 1.09, 95% CI 0.97 to 1.22), and its increment in C-index over age and stage was between −0.022 and +0.018. dMMR was classified with AUC 0.958, but random gene sets reached a median near 0.94 (empirical p = 0.234). The panel predicted GPX4 dependence in 59 colorectal cell lines with AUC 0.760, again at the random-set median (0.776). Only one of five response models, oxaliplatin IC_{50}, outperformed random sets nominally (p = 0.045), and not after multiplicity correction (q = 0.22). In a metric-level comparison with ferroptosis signatures published since 2023, whose external 5-year AUCs fell to 0.63–0.65 from 0.74–0.97 in discovery cohorts, the frozen model reached 0.553 in GSE39582, consistent with a pervasive optimism gap. These findings show that, for this panel, apparent predictive performance does not demonstrate ferroptosis-specific information, and they argue for random-gene-set benchmarking as a default control in signature studies.

Keywords: Ferroptosis; colorectal cancer; prognostic signature; random gene sets; external validation; drug response; reproducibility.

1 Introduction

Colorectal cancer remains a leading cause of cancer-related mortality worldwide, and molecular markers that improve prognosis or treatment selection beyond stage are of continuing interest. Ferroptosis is a form of regulated cell death driven by iron-dependent lipid peroxidation and restrained by the glutathione–GPX4 axis and parallel suppressor pathways [1]. Genes in iron handling, polyunsaturated lipid metabolism and antioxidant defence have therefore been proposed as features for prognostic models and as determinants of drug sensitivity. Cell state, notably a mesenchymal phenotype, has been linked to GPX4 dependence [2, 3].

A recurring weakness in gene-signature studies is the lack of an appropriate null. A model built on any gene set of reasonable size may show non-trivial survival association, because proliferation, stromal content and tumour stage affect the expression of thousands of genes [16]. Without comparison with random gene sets processed identically, a modest C-index or AUC cannot be attributed to the biology that motivated the gene list.

This work develops an end-to-end computational pipeline for ferroptosis-related analysis in colon cancer in which that comparison is built in. The objectives were:

1. to train a gene-expression survival model in TCGA-COAD and freeze it before testing in independent cohorts;
2. to assess whether the panel carries information about mismatch-repair status and defines survival-relevant subtypes;
3. to assess whether the panel predicts GPX4 dependence

and drug response in colorectal cell lines;
4. to judge every result against equally sized random gene sets and against positive controls that show whether the machinery can detect a real signal; and
5. to benchmark the discrimination of the frozen model against published ferroptosis signatures using matched quantitative metrics.

The main contributions are: (i) a frozen-model, multi-cohort validation framework with explicit de-duplication of overlapping GEO series; (ii) a random-gene-set null applied uniformly to survival, classification, subtype and drug-response tasks; and (iii) a metric-level comparison with signatures published from 2023 onwards that reports both discovery and external performance and quantifies the optimism gap.

2 Related Work

Numerous studies have reported ferroptosis-related signatures with prognostic value in colorectal and other cancers, typically derived by univariate screening followed by penalised Cox regression. Performance is often reported in the discovery cohort or in a single validation set, and rarely against random-gene-set baselines. Methodological work has shown that many random gene sets are associated with outcome [16], and that feature selection outside the cross-validation loop inflates performance. Large pharmacogenomic resources [5, 6] and the DepMap project [7] enable cell-line analyses, but the correspondence between cell-line response and patient outcome is limited.

Since 2023, compact colorectal models have continued to appear, including four- to ten-gene LASSO Cox scores validated in GSE39582 [13, 18, 19], multi-model machine-learning studies [12, 17], and signatures combining ferroptosis with cuproptosis or metabolic programmes [19, 20]. A 2026 pan-cancer benchmark is a notable exception, because it showed that an integrated cell-death signature exceeded size-matched random gene sets but did not outperform a ferroptosis-only model [22]. The present study therefore emphasises out-of-fold estimation, frozen models, empirical comparison with random sets, and explicit reporting of negative results. A quantitative comparison with the published models is given in Section 4.10.

3 Materials and Methods

3.1 Overview

Figure 1 summarises the workflow. The code is a single Python script (version 1.5.0, fixed seed 42) that downloads data, preprocesses them, runs all analyses and writes tables, figures and a provenance manifest. Nothing is simulated: an analysis whose data cannot be retrieved is skipped and reported as such. Figure 2 shows how each score and classifier is computed from the standardised panel expression.

3.2 Datasets

Table 1 lists the data used. TCGA-COAD expression (STAR gene-level counts, GDC Data Release 46.0) was converted to $\log_2(\mathrm{TPM}+1)$ for protein-coding genes. One primary-tumour sample per patient was kept (524 files reduced to 458 patients).

Table 1: Cohorts after quality control. Events are deaths (OS) or recurrences (DFS).

Cohort	Type	Endpoint	n	Events
TCGA-COAD	RNA-seq	OS	434	95
GSE39582	GPL570	OS	573	190
GSE17536	GPL570	OS	177	73
GSE14333	GPL570	DFS	153	29
DepMap/GDSC2	cell lines	see text	39–59	–

Overall survival (OS) was defined from vital status and days to death or last follow-up, with 434 patients valid (time > 0, event known; 95 deaths). The GEO cohorts (GSE39582 [4], GSE17536, GSE14333) are Affymetrix GPL570 microarrays; probes were collapsed to genes by retaining the probe with the highest mean expression and discarding probes mapping to several genes. GSE14333 provides no OS, so disease-free survival (DFS) was used (exploratory, endpoint mismatch with the OS-trained model). Pairwise correlation of samples between GEO cohorts identified 95 GSE14333 samples that duplicate GSE17536 ($r > 0.9$); they were removed, leaving 153 patients with DFS and 29 events.

For cell-line analyses, expression was taken from DepMap (protein-coding TPM, $\log_2(\mathrm{TPM}+1)$). The download of the GDSC RMA expression file and of CTRPv2 failed in the final run, so CTRP was not analysed; GDSC2 fitted dose–response values (colorectal lineage) were matched to DepMap lines by COSMIC identifier or normalised name. Genetic GPX4 dependence was taken from the DepMap CRISPR gene-effect score for 59 colorectal (bowel-lineage) lines with expression and score. No GPX4-inhibitor (ML162, ML210, RSL3) or erastin data were present in the retrieved GDSC2 file.

3.3 Ferroptosis Gene Panel

The panel comprises 36 genes (16 drivers, 20 suppressors; Appendix Table 12) compiled from the general ferroptosis literature. It is a candidate list that has not been independently curated against a ferroptosis database. All findings are therefore conditional on this panel (hash 03d5812616df).

3.4 Preprocessing and Harmonisation

RNA-seq and microarray values are never pooled. In each external cohort the selected genes are z-scored within the cohort using no outcome information and multiplied by coefficients frozen in TCGA. No validation outcome is used for selection, tuning or thresholds.

3.5 Survival Risk Model

The Cox model [10] is

$$h(t;\mathbf{x}) = h_0(t)\exp\left(\boldsymbol{\beta}^T\mathbf{x}\right), \tag{1}$$

and the risk score for patient i is

$$R_i = \sum_{j=1}^{p}\beta_j z_{ij}, \tag{2}$$

where z_{ij} is the standardised expression of selected gene j. In TCGA, genes are standardised, screened by univariate Cox

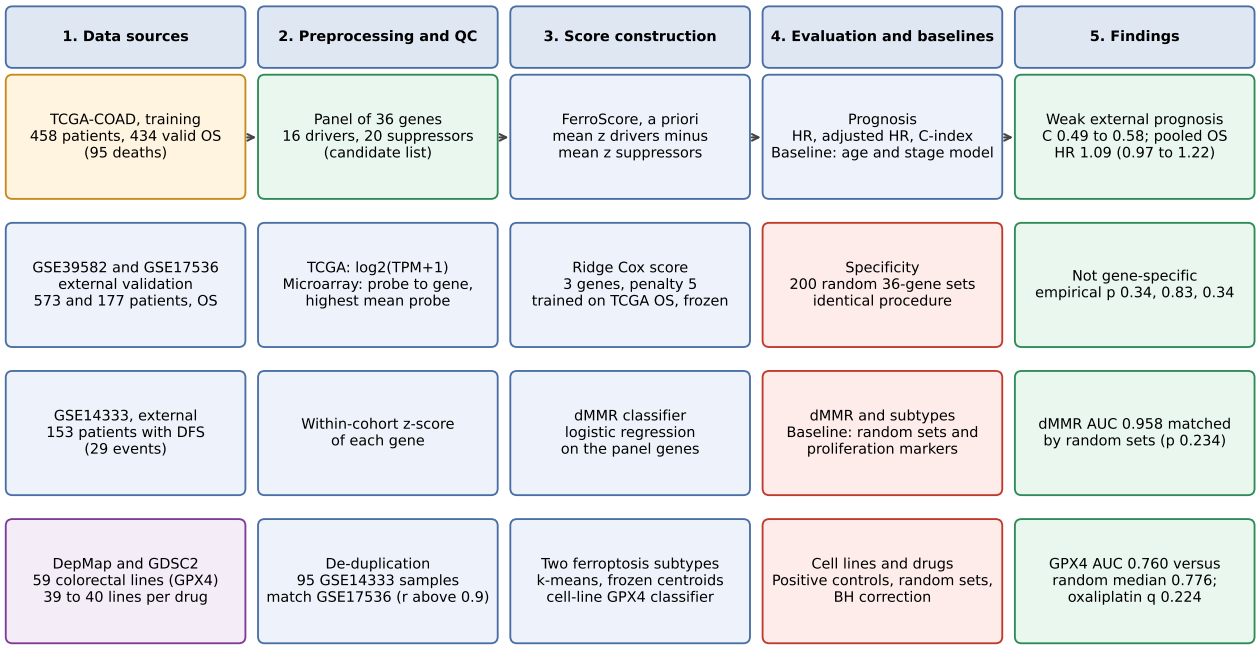


Figure 1: Architecture of the study. Public colorectal cancer and cell-line data are harmonised (stages 1 and 2), the scores and classifiers are constructed (stage 3), and each evaluation module is paired with a baseline or null model (stage 4, shaded red) before the findings are drawn (stage 5). Counts are those retained after quality control.

($p < 0.10$), and a ridge-penalised Cox model is fitted with the penalty chosen from $\{0.05, 0.5, 5, 50, 500\}$ by 3-fold inner cross-validation of the C-index. Three genes were selected (ALOX12, AKR1C1, HSPB1) with penalty 5. The model and the median risk cut-off were frozen before external evaluation. TCGA cross-validation (5-fold, 2 repeats) repeats screening, tuning and fitting inside each training fold. A clinical model (age, stage) and a combined model (age, stage, risk) were fitted in TCGA and frozen likewise. Discrimination is Harrell's C-index [8] with bootstrap 95% confidence intervals (500 resamples).

3.6 Random-Gene-Set Benchmark

For every task, 200 random gene sets of the same size as the panel (36 genes) were drawn without replacement from expressed genes shared by all cohorts, excluding panel genes. Each set passes through the *identical* procedure (screening, tuning, fitting, frozen application). The empirical one-sided p-value is

$$p_{\text{emp}} = \frac{1 + \#\{c_k \geq c_{\text{obs}}\}}{1 + N}, \tag{3}$$

with $N = 200$ and c the relevant metric. For subtype survival tests, the absolute log hazard ratio is compared. A p_{emp} near or above 0.05 means the panel cannot be distinguished from arbitrary genes for that task. Because p_{emp} cannot fall below one in 201, it is a resolution-limited statistic.

3.7 A Priori FerroScore

To test the ferroptosis genes without any fitting, a score was defined from the roles assigned to the panel genes before any survival analysis:

$$\text{FerroScore}_i = \frac{1}{n_D}\sum_{j\in D} z_{ij} - \frac{1}{n_S}\sum_{j\in S} z_{ij}, \qquad z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \tag{4}$$

where D and S are the 16 driver and 20 suppressor genes ($n_D = 16$, $n_S = 20$), and μ_j and σ_j are the mean and standard deviation of gene j within the cohort. The driver mean and the suppressor mean were also analysed separately. This component analysis was added after the main results were seen and is exploratory.

3.8 Molecular Classification

dMMR (1) versus pMMR (0) labels were parsed from GSE39582 (536 patients with labels, 77 dMMR, 14.4%). An L2 logistic regression,

$$P(Y = 1; \mathbf{x}) = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})]}, \tag{5}$$

was evaluated by patient-level out-of-fold (OOF) prediction (5-fold, 2 repeats) with in-fold standardisation. A 12-gene proliferation marker set served as a comparator. TCGA MSI labels were not retrieved.

3.9 Ferroptosis Subtypes

k-means with $k = 2$ (prespecified) was fitted on TCGA z-scores of the panel; the cluster with the higher mean driver-minus-suppressor score was labelled "high". Centroids were frozen, and each GEO cohort was assigned by nearest centroid after cohort-wise z-scoring. Subtypes were tested against

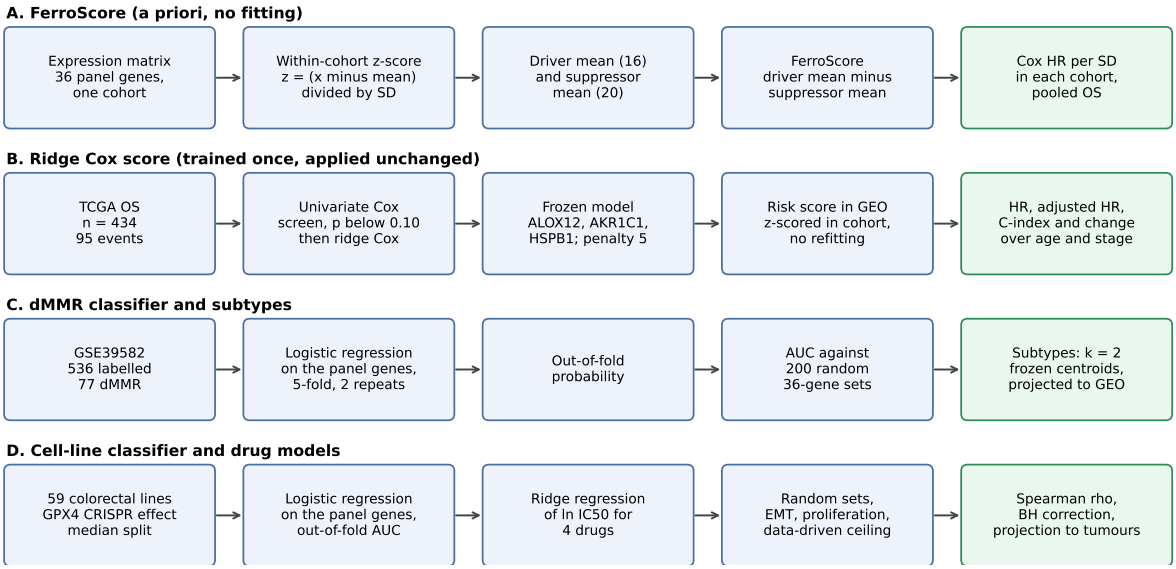


Figure 2: Method-level architecture. After within-cohort standardisation the panel feeds (A) the a priori FerroScore, (B) the ridge Cox score trained once on TCGA overall survival and applied unchanged to the GEO cohorts, (C) the dMMR classifier and the ferroptosis subtypes, and (D) the cell-line classifier and drug models. Green boxes are the outputs that are judged against baselines.

survival (log-rank, Cox with and without age and stage), stage, age and MMR, and compared with random sets by the same clustering-and-projection procedure.

3.10 Cell-Line Sensitivity and Drug-Response Prediction

Colorectal cell lines were labelled sensitive or resistant to GPX4 loss by the median split of the CRISPR gene-effect score (29 sensitive, 30 resistant). The panel-based logistic classifier was evaluated by OOF AUC (5-fold, 3 repeats), a label-permutation test (200 permutations) and the random-set benchmark. The frozen classifier was then projected onto the tumour cohorts. For dose–response, ridge regression

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left[\sum_{i=1}^{n} (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2 + \lambda \sum_j \beta_j^2 \right] \tag{6}$$

predicted ln IC_{50} (μM) of 5-fluorouracil, irinotecan, oxaliplatin and SN-38, and the CRISPR gene effect, from panel expression (y_i the response of line i). Performance is the Spearman correlation ρ between OOF predictions and measurements, judged against random sets, with Benjamini–Hochberg (BH) [9] correction across the five models. Out-of-fold ρ is biased low for pure noise, so only the comparison with random sets is interpreted.

3.11 Positive Controls

To test whether the machinery can detect real signal, three references were run identically: a mesenchymal and epithelial (EMT) signature (24 genes) and a proliferation signature (12 genes) for GPX4 dependence and survival; clinical stage for survival; and a data-driven ceiling in which the 36 most discriminative genes are chosen inside each training fold. Cell-line results describe in vitro response and are not clinical doses.

3.12 Adjustment, Incremental Value and Pooling

Within each external cohort the frozen risk score was standardised, and the hazard ratio per standard deviation (SD) was estimated by Cox regression. Adjusted models included ordinal stage (TNM or AJCC stage in GSE39582 and GSE17536, Dukes A to D coded 1 to 4 in GSE14333) and age per 10 years, using complete cases. The proportional-hazards assumption was tested with scaled Schoenfeld residuals. Incremental discrimination was defined as

$$\Delta C = C_{\text{age+stage+risk}} - C_{\text{age+stage}}, \tag{7}$$

where both models carry coefficients frozen in TCGA, and the 95% interval was obtained from 1,000 bootstrap resamples of the same patients. Log-rank tests used the frozen TCGA median of the risk score, and risk tertiles used the frozen TCGA tertile cut-offs. Cohort-level log hazard ratios were combined by inverse-variance fixed-effect and DerSimonian–Laird random-effects models, and heterogeneity was summarised by I^2. Pooling was done within the overall-survival endpoint, because mixing endpoints is not interpretable.

3.13 Metric-Level Benchmarking Against Published Signatures

Published ferroptosis signatures serve as external reference points through a fixed protocol based on reported performance metrics.

Study selection. Studies were included if they were published from 2023 onwards, built a ferroptosis-related gene model for colon or colorectal cancer from TCGA and GEO data, and reported at least one discrimination metric. One pan-cancer benchmark that applied a random-gene null was added because it tests the same question as this study. The search was targeted and not systematic.

Metric extraction. For every study the following were recorded: signature size, discovery cohort, external cohort, evaluation horizon, and the reported time-dependent AUC or C-index. Values were read from abstracts, methods text and result summaries. Published models were not refitted, because gene lists, coefficients and preprocessing differ.

Metric alignment. Metrics of the same type are compared directly (AUC with AUC, C-index with C-index) and are never ranked across types. Because published studies mostly report time-dependent ROC AUC, whereas the pipeline reports Harrell's C, the inverse-probability-of-censoring-weighted cumulative and dynamic AUC (Uno-type estimator) was computed for the frozen risk score at 36 and 60 months, with 300 bootstrap resamples for intervals. Estimator details differ between studies, so differences of a few hundredths are not interpreted.

Optimism gap. For each study with both discovery and external values at the same horizon, the optimism gap is

$$\Delta_{\text{opt}}(\tau) = \text{AUC}_{\text{disc}}(\tau) - \text{AUC}_{\text{ext}}(\tau), \tag{8}$$

which measures how much of the discovery performance is lost on independent data. For this study, AUC_{disc} is the TCGA in-sample AUC of the frozen model and AUC_{ext} is the GSE39582 AUC.

Comparability grading. Each pair is graded *high* when the metric type and external cohort coincide, *moderate* when the metric type is the same but the cohort or cancer scope differs or both values are in-sample, and *low* when the metric type differs or the published value belongs to a model combined with clinical variables.

4 Results

4.1 Cohort Characteristics

After quality control the four tumour cohorts contained 434, 573, 177 and 153 evaluable patients (Table 1). In TCGA, median follow-up was 22.1 months, median age 68.5 years, and 19.4% of stage values were missing. The panel was fully present in all cohorts (36 of 36 genes).

4.2 Survival Model Performance

Univariate screening in TCGA retained ALOX12 (HR 1.40 per SD, 95% CI 1.18–1.65, BH $q = 0.005$), AKR1C1 (HR 1.23) and HSPB1 (HR 1.23); only ALOX12 survived multiplicity correction. TCGA cross-validation with selection inside folds gave $C = 0.548$ for the molecular model, against 0.770 for the clinical model and 0.774 for the combined model (in-sample molecular $C = 0.617$, optimistic). Table 2 gives the frozen external results. The molecular model was close to chance in all cohorts, and the clinical model was clearly superior in the two OS cohorts (Fig. 3; risk-group survival curves are shown in Fig. 4). Adding the molecular risk score to the clinical model changed C from 0.675 to 0.683 in GSE39582 and from 0.747 to 0.725 in GSE17536 (Fig. 11).

Table 2: Frozen external validation (Harrell C, 95% bootstrap CI) and comparison with 200 random gene sets.

Cohort	Molecular C	Random median	p_{emp}	Clinical C
GSE39582	0.539 [0.499, 0.580]	0.528	0.338	0.675
GSE17536	0.486 [0.415, 0.562]	0.532	0.831	0.747
GSE14333[†]	0.581 [0.471, 0.701]	0.555	0.343	0.624

[†]DFS endpoint, 29 events; exploratory.

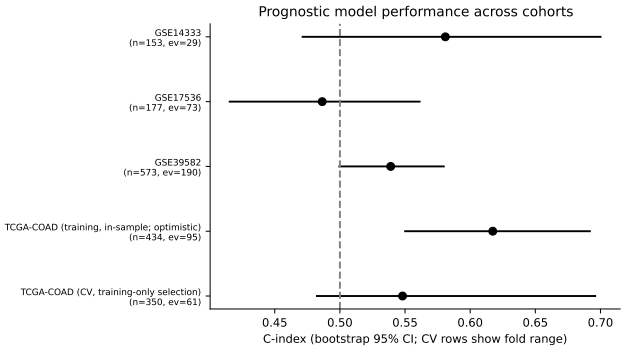


Figure 3: Concordance index of the molecular model: TCGA cross-validation (training-only selection, interval shows fold range), TCGA in-sample, and frozen external cohorts.

4.3 Adjusted Association and Incremental Value

Unadjusted, the frozen score was not significantly associated with outcome in any external cohort (Table 3, Fig. 5). The hazard ratio per SD was 1.13 (95% CI 0.99 to 1.29, $p = 0.080$) for OS in GSE39582, 0.97 (0.76 to 1.23, $p = 0.783$) for OS in GSE17536 and 1.34 (0.93 to 1.93, $p = 0.120$) for DFS in GSE14333. The proportional-hazards assumption was not rejected in any cohort (Schoenfeld p between 0.29 and 0.74). The fixed-effect pooled OS estimate was 1.09 (0.97 to 1.22, $p = 0.164$), with low heterogeneity ($I^2 = 0.17$).

After adjustment for age and stage, the estimate became significant in GSE39582 (HR 1.20, 1.04 to 1.37, $p = 0.011$) but reversed direction in GSE17536 (HR 0.75, 0.58 to 0.98, $p = 0.035$), and remained non-significant for DFS in GSE14333 (HR 1.24, 0.87 to 1.76, $p = 0.230$). The two OS cohorts therefore disagree after adjustment ($I^2 = 0.89$), the random-effects pooled estimate was 0.96 (0.61 to 1.51), and the adjusted fixed-effect pooled value (1.08, 0.96 to 1.22) should not be interpreted. Because the unadjusted estimate in GSE17536 is close to 1 while stage is strongly prognostic there, the reversal is most likely a statistical consequence of conditioning on stage in a cohort of 177 patients, and not evidence of a protective effect.

Incremental discrimination over age and stage was small or negative (Table 4). The change in C-index was +0.008 (95% bootstrap interval −0.004 to +0.022) in GSE39582, −0.022 (−0.039 to −0.003) in GSE17536, where adding the score lowered discrimination, and +0.018 (−0.018 to +0.057) in GSE14333. In GSE39582, the frozen TCGA tertiles gave a monotonic but marginal pattern: relative to the lowest tertile, the middle tertile had HR 1.41 (0.98 to 2.02, $p = 0.061$) and the highest tertile 1.47 (1.00 to 2.15, $p = 0.047$). The continuous score remains the primary analysis, and the tertile result is exploratory.

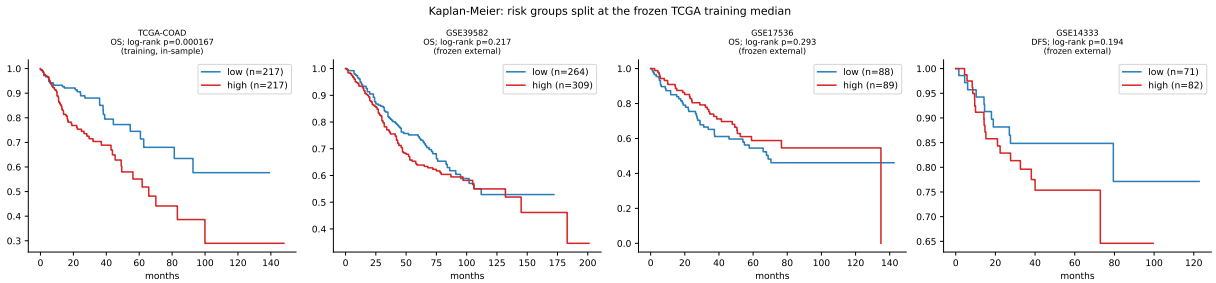


Figure 4: Kaplan–Meier curves for risk groups split at the frozen TCGA median risk, in the training cohort (in-sample) and the external cohorts.

Table 3: External association of the frozen three-gene score with survival, unadjusted and adjusted for age (per 10 years) and stage. Hazard ratios (HR) are per SD of the score within each cohort. PH p is the Schoenfeld-residual test of the proportional-hazards assumption. Adjusted models use complete cases (n = 568, 177 and 153).

Cohort	Endpoint	n	Events	Unadjusted HR (95% CI)	p	PH p	Adjusted HR (95% CI)	p
GSE39582	OS	573	190	1.13 (0.99 to 1.29)	0.080	0.29	1.20 (1.04 to 1.37)	0.011
GSE17536	OS	177	73	0.97 (0.76 to 1.23)	0.783	0.74	0.75 (0.58 to 0.98)	0.035
GSE14333	DFS	153	29	1.34 (0.93 to 1.93)	0.120	0.55	1.24 (0.87 to 1.76)	0.230

Table 4: Discrimination of the frozen score alone, of the frozen age and stage model, and of the combined model (Harrell C, complete cases), with the increment ΔC (bootstrap 95% interval) and the log-rank p for the median split.

Cohort	End.	Score	Age+stage	Combined	ΔC (95% CI)	Log-rank p
GSE39582	OS	0.537	0.675	0.683	+0.008 (−0.004 to +0.022)	0.217
GSE17536	OS	0.486	0.747	0.725	−0.022 (−0.039 to −0.003)	0.293
GSE14333	DFS	0.581	0.624	0.642	+0.018 (−0.018 to +0.057)	0.194

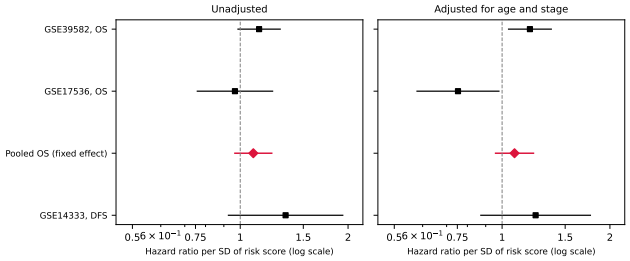


Figure 5: Hazard ratio per SD of the frozen score in the external cohorts. Squares are single cohorts and the red diamond is the fixed-effect pooled estimate for OS. Left, unadjusted; right, adjusted for age and stage.

4.4 A Priori FerroScore and Its Components

The a priori FerroScore, which involves no fitting, was not associated with survival in any cohort (Table 5). Hazard ratios per SD were 1.07 in TCGA-COAD, 0.93 in GSE39582, 0.88 in GSE17536 and 1.05 for DFS in GSE14333, with a fixed-effect pooled OS estimate of 0.95 (0.86 to 1.05, $p = 0.331$, $I^2 = 0$). In the exploratory component analysis, the only nominally significant result was a protective association of the driver mean in GSE39582 (HR 0.80, 0.70 to 0.93, $p = 0.003$). It was not replicated in TCGA-COAD (1.13) or GSE17536 (1.03), the pooled OS estimate was 0.92 (0.83 to 1.03) with $I^2 = 0.76$, and it is one of twelve unadjusted tests. The suppressor mean had a pooled OS hazard ratio of 0.99 (0.90 to 1.10). The ferroptosis role assignment therefore carries no demonstrable prognostic information in these cohorts.

4.5 Specificity Against Random Gene Sets

In none of the three external cohorts did the panel outperform random sets (Fig. 6, Table 2); the 95th percentile of random-set C was 0.575, 0.594 and 0.646, above the observed value in each. The proliferation marker set reached $C = 0.570$ in GSE39582, but 0.498 and 0.509 in the other cohorts, so no gene set was consistently prognostic.

4.6 Classification of dMMR

The panel classified dMMR with OOF AUC 0.958 (95% CI 0.932–0.976; balanced accuracy 0.884; recall 0.857; specificity 0.911; precision 0.617 at prevalence 14.4%) (Fig. 7). The proliferation set reached 0.934 (0.899–0.962). Random gene sets, however, achieved similarly high AUCs, and the panel was not distinguishable from them ($p_{\text{emp}} = 0.234$). The unsupervised driver-minus-suppressor score had AUC 0.453, i.e. no informative orientation. The high AUC therefore reflects the strong, genome-wide expression signature of dMMR rather than ferroptosis biology.

4.7 Ferroptosis Subtypes

The two-cluster solution was only moderately stable (bootstrap adjusted Rand index 0.55; silhouette 0.19 for $k = 2$, declining for larger k). High-ferroptosis tumours showed no significant survival difference in TCGA (HR 0.87, log-rank $p = 0.643$), GSE39582 (HR 1.11, $p = 0.637$), GSE17536 (HR 0.52, $p = 0.099$) or GSE14333 (HR 0.45, $p = 0.424$) (Fig. 8); adjustment for age and stage did not change this. Compared with random sets, the empirical p-values for the absolute

Table 5: A priori FerroScore and its components (HR per SD, 95% CI, p) in the tumour cohorts. TCGA-COAD values are not external. The component analysis is exploratory.

Cohort	Endpoint	n	FerroScore	Driver mean	Suppressor mean
TCGA-COAD	OS	434	1.07 (0.88 to 1.30), 0.516	1.13 (0.92 to 1.38), 0.243	1.05 (0.86 to 1.29), 0.632
GSE39582	OS	573	0.93 (0.81 to 1.07), 0.317	0.80 (0.70 to 0.93), 0.003	0.92 (0.80 to 1.06), 0.229
GSE17536	OS	177	0.88 (0.71 to 1.08), 0.220	1.03 (0.81 to 1.29), 0.825	1.14 (0.91 to 1.43), 0.239
GSE14333	DFS	153	1.05 (0.73 to 1.52), 0.785	1.06 (0.74 to 1.53), 0.751	0.99 (0.69 to 1.44), 0.977
Pooled OS, fixed effect	OS	3 cohorts	0.95 (0.86 to 1.05)	0.92 (0.83 to 1.03)	0.99 (0.90 to 1.10)

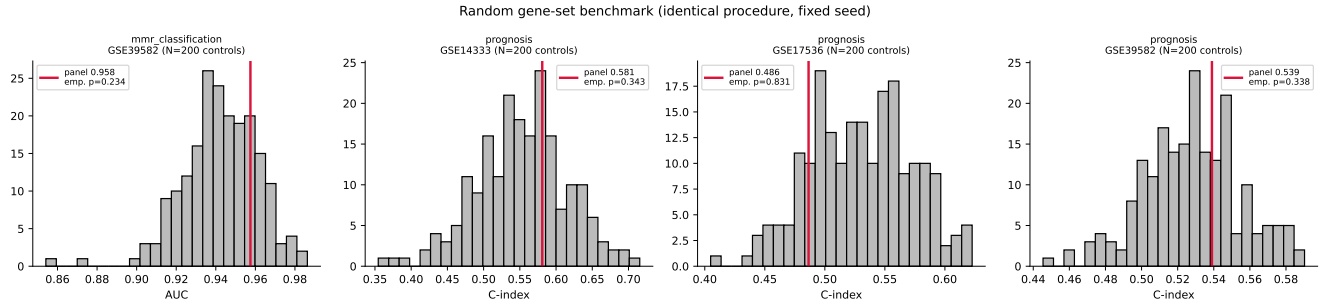


Figure 6: Panel performance (red line) versus the distribution of 200 random gene sets processed by the identical pipeline: dMMR classification (AUC) and frozen prognosis (C-index) in the three external cohorts.

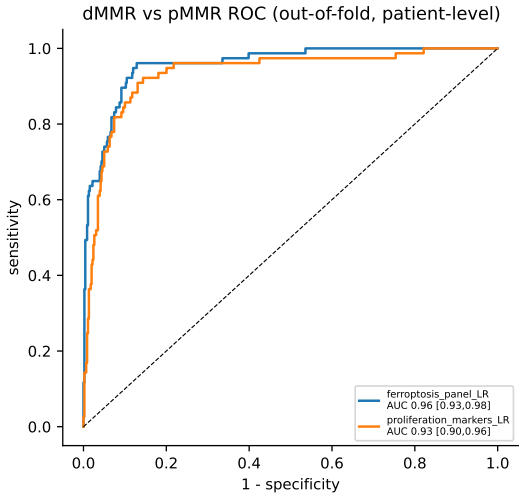


Figure 7: Out-of-fold ROC curves for dMMR vs. pMMR in GSE39582 (panel and proliferation markers).

Table 6: Positive controls for GPX4 dependence (59 lines, OOF AUC) against 200 random sets.

Signature	Genes	AUC	Random median	p_{emp}
Ferroptosis panel	36	0.760	0.776	0.637
EMT	24	0.590	0.755	0.980
Proliferation	12	0.671	0.722	0.771
Data-driven ceiling	36	0.818	–	–

Table 7: Ridge prediction of cell-line response from the panel (OOF Spearman ρ) versus random gene sets.

Response	n	ρ	Random med.	p_{emp}	q_{BH}
5-Fluorouracil	40	0.127	0.183	0.647	0.809
Irinotecan	40	−0.031	−0.108	0.388	0.809
Oxaliplatin	40	0.480	0.201	0.045	0.224
SN-38	39	−0.171	−0.146	0.557	0.809
GPX4 CRISPR effect	59	−0.054	0.334	0.985	0.985

log hazard ratio were 0.42, 0.64, 0.37 and 0.20. The high subtype contained fewer dMMR tumours in GSE39582 (3.1% vs. 15.9%, Fisher $p = 0.004$), but this difference was not larger than for random sets ($p_{\text{emp}} = 0.259$). No association with stage or age was found.

4.8 Cell-Line Sensitivity and Drug-Response Analysis

The panel predicted GPX4 dependence in 59 colorectal lines with OOF AUC 0.760 (95% CI 0.645–0.887; permutation $p = 0.010$), so the panel does contain information about the label, but random sets achieved a median AUC of 0.776 ($p_{\text{emp}} = 0.637$). The positive controls clarify the limits (Table 6, Fig. 9b): the EMT signature (AUC 0.590) and the proliferation signature (0.671) did not recover the expected signal, whereas selecting 36 genes inside folds gave a ceiling of 0.818. Hence, at $n = 59$, the benchmark has little power to separate a specific panel from random sets.

Projection of the frozen classifier onto tumours showed no association with survival in any cohort (HR per SD 0.99–1.24; all $p > 0.1$; p_{emp} 0.54–0.95), although the projected score correlated weakly with dMMR in GSE39582 (AUC 0.406, Mann–Whitney $p = 0.008$).

For chemotherapy response (Table 7), only oxaliplatin showed a correlation between predicted and measured ln IC_{50} exceeding random sets at the nominal level ($\rho = 0.480$, 95% CI 0.183–0.702; $p_{\text{emp}} = 0.045$), which is not significant after BH correction ($q = 0.224$). The other four models were not distinguishable from random sets. With five tests, and with the lower bound of p_{emp} limited by 200 sets, this should be regarded as a hypothesis for replication, not a finding.

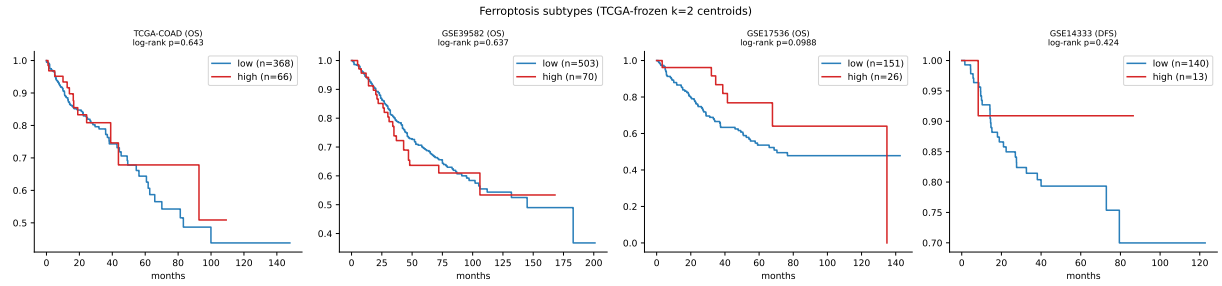


Figure 8: Kaplan–Meier curves for the two ferroptosis subtypes defined by TCGA-frozen centroids.

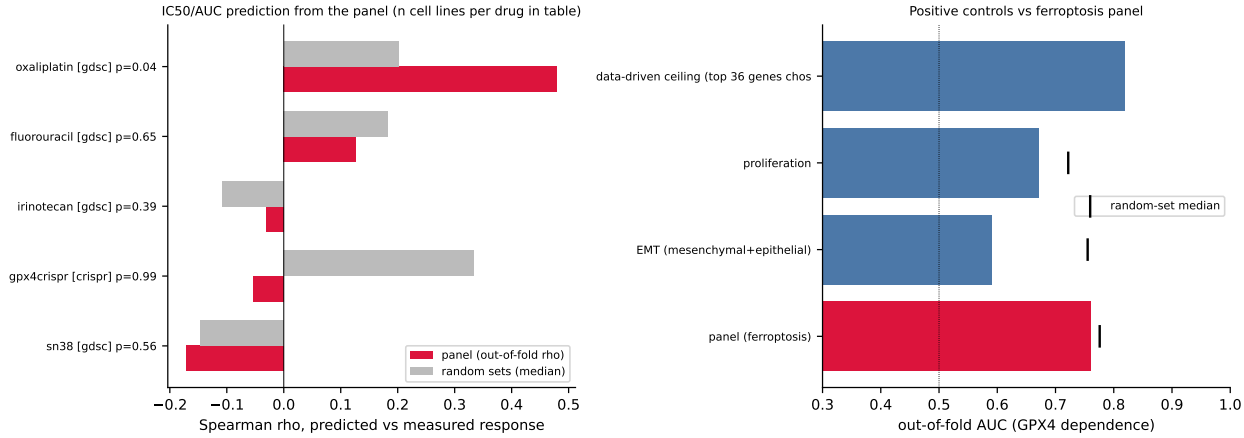


Figure 9: (a) Out-of-fold correlation between predicted and measured response for the panel (red) and median of random sets (grey). (b) Positive controls and ferroptosis panel for prediction of GPX4 dependence; black ticks mark random-set medians.

4.9 Sensitivity Analyses

Clinical stage was strongly prognostic in all cohorts (HR per SD 1.72–2.77), confirming that the survival machinery detects real signal. In contrast, proliferation (HR per SD 0.81–1.05) and EMT scores (1.10–1.22) had weak or inconsistent effects; EMT was nominally associated with OS in GSE39582 (HR 1.22, $p = 0.004$), but not larger than random sets ($p_{\text{emp}} = 0.075$).

4.10 Benchmarking Against Published Signatures

The comparison follows the metric-level protocol of Section 3.13. Table 8 lists the time-dependent AUCs reported by each published colorectal ferroptosis model in its discovery and external settings, together with the frozen model of this study evaluated by the same metric, and Table 9 lists C-index-type metrics. Table 10 gives the optimism gap for every study that reports both values at the same horizon.

In GSE39582, the one external cohort shared with a published model, the 3-year and 5-year AUCs of 0.62 and 0.63 reported by Wang [13] exceed the frozen model (0.557 and 0.553) by 0.06 to 0.08, and the bootstrap interval of the present 5-year estimate (0.499 to 0.614) lies at or below the published values (comparability high). Against Gan [12], whose external AUC at 5 years was 0.649 in GSE29621, the difference is 0.10 (comparability moderate, different cohort). Published external AUCs therefore span 0.59 to 0.67 across horizons, and the frozen model lies at the lower edge of that range.

The optimism gap separates the models more clearly than

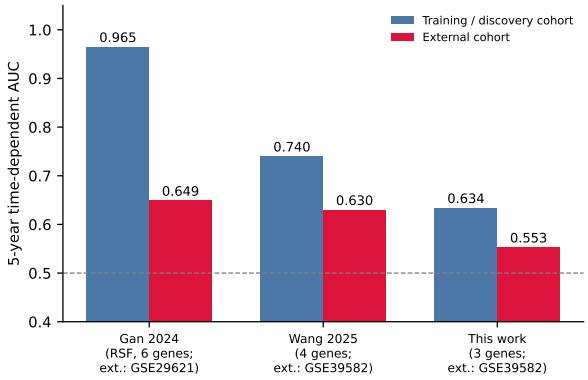


Figure 10: Five-year time-dependent AUC in discovery and external cohorts for two published ferroptosis models and this study. The dashed line marks chance. For this study the discovery value is the TCGA in-sample AUC (optimistic) and the external value is GSE39582.

the external values (Table 10). The random survival forest lost 0.32 AUC at 5 years between discovery and external data, and the LASSO Cox model of Wang lost 0.11. The frozen model of this study lost 0.08 even though its in-sample AUC (0.634) was modest, which is consistent with a thin signal that was fitted to TCGA and did not transport. The in-sample TCGA AUCs of this study (0.616 and 0.634) are also lower than the training AUCs reported for the 2026 model (0.728 and 0.808 at 3 and 5 years), both of which are optimistic (comparability moderate).

For C-index metrics (Table 9), the external C of the frozen

Table 8: Time-dependent AUC of published ferroptosis-related colorectal models and of the frozen model of this study. Discovery values for published models are those reported in the original study; for this study the discovery value is the TCGA in-sample AUC (optimistic). n.r.: not reported in the text reviewed. Horizons are in years.

Model	Genes	Method	Discovery AUC (horizon)	External AUC (cohort, horizon)	Note
Guo 2023 [11]	39	–	0.81 (2), 0.81 (5), 0.78 (8)	n.r.	Split not stated[a]
Li 2023 [14]	3	–	n.r.	0.568 (GEO)	Second-hand value[b]
Wang 2023 [17]	23	Lasso, plsRcox	n.r.	n.r.	Best of 83 algorithm combinations
Zeng 2024 [18]	4	LASSO Cox	0.897 (TCGA)	0.858 (GSE39582)	Score plus 3 clinical factors
Gan 2024 [12]	6	Random survival forest	0.978 (1), 0.985 (3), 0.965 (5)	0.671 (1), 0.589 (3), 0.649 (5) (GSE29621)	Discovery is TCGA-COAD cross-validation
Wang 2025 [13]	4	LASSO Cox	0.71 (3), 0.75 (4), 0.74 (5)	0.62 (3), 0.64 (4), 0.63 (5) (GSE39582)	External cohort of 585 samples
Gao 2026 [21]	8	–	0.857 (1), 0.728 (3), 0.808 (5)	n.r.	Training values only
This study	3	Ridge Cox	0.616 (3), 0.634 (5)	0.557 (3), 0.553 (5) (GSE39582)	95% CI at 5 years 0.499 to 0.614

[a] Abstract only. [b] Value re-evaluated in [13], not verified in the primary article. Values were taken from abstracts, methods text and result summaries, not re-analysed.

Table 9: *C*-index-type metrics. Values of different types are listed side by side but not ranked.

Source	Metric	Value
Demir and Yasar [22]	Pan-cancer *C*	0.573 (0.552 to 0.594)
Wang 2025 [13]	Nomogram *C*	0.78
This study, score alone	External *C*	0.539, 0.486, 0.581
This study, age + stage	External *C*	0.675, 0.747, 0.624
This study, combined	External *C*	0.683, 0.725, 0.642

Table 10: Optimism gap $\Delta_{opt} = AUC_{disc} - AUC_{ext}$ at matched horizons, with comparability of the external cohort to this study.

Study	1 y	3 y	5 y	Comparability
Gan 2024	0.307	0.396	0.316	Moderate
Wang 2025	–	0.090	0.110	High
This study	n.r.	0.059	0.081	–

Wang 2025 also reports 4-year AUCs of 0.75 (discovery) and 0.64 (external), a gap of 0.110.

score (0.486 to 0.581) brackets the pan-cancer value of 0.573 reported by Demir and Yasar [22]. That study is the only one in the comparison that applied a random-gene null; its integrated signature exceeded size-matched random sets, whereas the ferroptosis panel here did not (empirical p 0.338, 0.831 and 0.343), although the cancer scope differs (comparability moderate). The combined-model values reported by Zeng [18] (AUC 0.858) and Wang [13] (nomogram *C* 0.78) include clinical variables and are not comparable to the score-only values here (comparability low); the corresponding combined model of this study reached *C* 0.683 and 0.725 in the two OS cohorts, of which the age and stage model contributed nearly all (0.675 and 0.747).

Among the published models, no colorectal study reports a random-gene null, so their external AUCs cannot be judged against chance-level gene sets. For orientation, the 95th percentile of random-set *C* in this study was 0.575 to 0.646 across the three external cohorts, although this is a different metric. Table 11 collects each result of this study beside its internal baseline.

Caveats. Cohorts, endpoints, gene-set definitions and AUC estimators differ between studies, so the comparison is indicative and not a formal meta-analysis. Published models were not refitted. Several entries were taken from abstracts, and the entries for Xu, Wang 2023, Guan, the cuproptosis and ferroptosis model and Gao were not re-verified against the full texts.

5 Discussion

A weak and non-replicated prognostic signal. The frozen three-gene score was not significantly associated with outcome in any single external cohort, and the pooled OS hazard ratio per SD of 1.09 (0.97 to 1.22) included 1. The adjusted estimates disagreed between the two OS cohorts, and the increment in *C*-index over age and stage ranged from −0.022 to +0.018. Discrimination of the score alone (0.49 to 0.58) was close to chance, whereas age and stage reached 0.62 to 0.75. The in-sample TCGA value (0.617) and the cross-validated value (0.548) differ by almost as much as the external values differ from chance, which is the usual signature of optimism in a discovery cohort. Only ALOX12 survived multiplicity correction in the TCGA screening, so the signal available for model building was thin from the start. Two of the three selected genes (AKR1C1 and ALOX12) also appear in a five-gene colon signature published in 2021 [15], which reached AUCs of only 0.546 to 0.578 at 1 to 3 years in its GEO test cohort, close to the external discrimination observed here.

No gene-specific information. In none of the three external cohorts did the panel outperform random gene sets of the same size, and the a priori FerroScore, which needs no fitting, had a pooled OS hazard ratio of 0.95 (0.86 to 1.05). The one nominal component result (driver mean in GSE39582) was not replicated and arises among twelve tests. These data therefore

Table 11: Summary of each result of this study against its internal baseline.

Question	Score or model	Baseline	Outcome
Prognosis beyond clinical data	Frozen molecular model, C 0.539, 0.486 and 0.581	Age and stage model, C 0.675, 0.747 and 0.624	Molecular model below the clinical model in all three cohorts
Specificity of prognosis	Empirical p 0.338, 0.831 and 0.343	200 random 36-gene sets, median C 0.528, 0.532 and 0.555	Not distinguishable from random sets
Survival of subtypes	High versus low subtype, log-rank p 0.643, 0.637, 0.099 and 0.424	200 random sets, empirical p 0.42, 0.64, 0.37 and 0.20	Not distinguishable from random sets
dMMR classification	Panel AUC 0.958	200 random sets (empirical p 0.234); proliferation markers AUC 0.934	Not distinguishable from random sets
GPX4 dependence in cell lines	Panel AUC 0.760, permutation p 0.010	Random-set median AUC 0.776 (empirical p 0.637); EMT 0.590; proliferation 0.671; ceiling 0.818	Signal present against label permutation, not specific to the panel
Oxaliplatin IC_{50}	Out-of-fold ρ 0.480	Random-set median ρ 0.201 (empirical p 0.045, BH q 0.224)	Nominal only; not significant after correction

do not support the idea that the annotated ferroptosis role of the genes carries prognostic information in colon cancer. The proliferation markers were prognostic in one cohort only ($C = 0.570$ in GSE39582, 0.498 and 0.509 elsewhere), so no gene set was consistently informative.

Mismatch repair status. An AUC of 0.958 for dMMR would conventionally be read as strong evidence. Random gene sets of equal size reached similar values ($p_{\text{emp}} = 0.234$), and the unsupervised driver-minus-suppressor score had AUC 0.453. The result reflects the broad transcriptomic footprint of mismatch-repair deficiency and not ferroptosis biology. The same applies to the ferroptosis subtypes, whose lower dMMR fraction in GSE39582 (3.1% against 15.9%) was not larger than for random sets.

Cell-line sensitivity and drug response. The panel predicted GPX4 dependence with an AUC of 0.760 that was significant against label permutation ($p = 0.010$) but unremarkable against random sets (median 0.776). A permutation test answers whether any signal exists, whereas the random-set test answers whether the signal belongs to the gene list, and only the second matters for a signature. The positive controls limit what can be concluded: the EMT signature (0.590) and the proliferation signature (0.671) did not recover the expected signal, while the in-fold ceiling reached 0.818, so the benchmark has limited power with 59 lines. Oxaliplatin was the only drug model that exceeded random sets nominally ($\rho = 0.480$, $p_{\text{emp}} = 0.045$), and it did not survive correction ($q = 0.224$). It is a hypothesis for replication and not a finding. Wang [13] likewise reported that the risk score did not separate predicted IC_{50} for 5-fluorouracil, irinotecan or oxaliplatin, in line with the absence of a robust drug-response signal here. Cell-line response is not a clinical dose.

Position relative to published signatures. On matched metrics, the external 5-year AUC of the frozen model (0.553) is 0.08 below the 0.63 reported by Wang in the same cohort and 0.10 below the 0.649 reported by Gan in a different cohort, whereas the discovery-to-external gap was the smallest of the three (0.08 versus 0.11 and 0.32). The pattern is that of a model with little signal to lose. The larger published AUCs were obtained in discovery cohorts (up to 0.985), and published external values fall to 0.59–0.67, which is the range in which a random-gene null becomes decisive. No colorectal study in the comparison reported such a null; the only study that did [22] reached the same qualitative conclusion for an integrated signature. The supported statement is therefore narrow: external discrimination is modest and of the same order as published external values, and, unlike in the colorectal studies, the share attributable to the specific genes is tested and found to be negligible.

Practical implications. The scores in this study do not support clinical use for prognosis beyond stage or for treatment selection. The results support random-signature benchmarking, frozen external validation, de-duplication of overlapping GEO cohorts and positive controls as routine practice, because two GEO series in this work shared 95 samples that would otherwise have inflated the evidence. Two checks show that the pipeline is able to detect signal when it exists: clinical stage was strongly prognostic in every cohort (hazard ratio per SD 1.72 to 2.77), and the data-driven ceiling in cell lines was clearly above the panel.

6 Limitations

The gene panel is a literature-derived candidate list that has not been curated against a ferroptosis database, and findings may differ for another panel. GSE14333 provides DFS only and has 29 events after de-duplication, so the third validation is exploratory. The adjusted analysis in GSE17536 rests on 177 patients and produced an estimate in the opposite direction, which shows the instability of small-cohort adjustment. The component analysis and the tertile analysis were added after the main results and are exploratory, and many secondary tests report nominal p values. Cell-line analyses use 39 to 59 lines, the retrieved GDSC2 file contained no GPX4 inhibitors so that GPX4 CRISPR dependence served as a surrogate, CTRP could not be retrieved, and the power to separate the panel from random sets is correspondingly limited, as the positive controls show. TCGA microsatellite status and DFS were not retrieved, so dMMR classification was run only in GSE39582, with labels parsed from GEO annotations. RNA-seq and microarray data are not pooled, and cohort-wise standardisation assumes comparable case mix. Empirical p values cannot fall below one in 201. Published signatures were not refitted, so the benchmark compares reported metrics, which differ in estimator, cohort and horizon, and several entries were taken from abstracts. No

adjustment for proliferation or stroma was made for the frozen score. Retrospective data carry selection bias and residual confounding, and no clinical application is supported without experimental and prospective validation.

7 Conclusion

This study presents a reproducible pipeline that pairs every ferroptosis-signature result with a baseline or null model. In colon cancer, a 36-gene ferroptosis panel did not outperform random gene sets for external survival prediction, dMMR classification, subtype survival association or GPX4-dependence prediction. The frozen three-gene score was not significantly associated with outcome in any single external cohort, added little or negative discrimination to age and stage, and the a priori FerroScore was null. In a metric-level comparison, the external 5-year AUC of 0.553 in GSE39582 lay below published external values of 0.63 to 0.65, while published discovery AUCs of 0.74 to 0.97 illustrate the optimism that a random-gene null is designed to expose. A possible oxaliplatin signal merits replication. Future work should use a curated panel, GPX4-inhibitor response data and larger cell-line panels, and should adjust for proliferation and stroma.

Declarations

Ethics approval: Not applicable; only public, de-identified data were analysed.

Data availability: TCGA-COAD (GDC), GEO accessions GSE39582, GSE17536 and GSE14333, GDSC2 and DepMap are public.

Code availability: `ferro_pipeline.py` v1.5.0; add repository URL before submission.

Funding: [Declare funding.]

Conflicts of interest: [Declare any competing interests.]

Author contributions: [Specify contributions.]

A Gene Panel

Table 12: The 36-gene candidate ferroptosis panel.

Role	Genes
Driver (16)	ACSL4, LPCAT3, ALOX15, ALOX12, ALOX5, POR, PEBP1, NCOA4, TFRC, HMOX1, SAT1, NOX1, TP53, GLS2, DPP4, BECN1
Suppressor (20)	GPX4, SLC7A11, AIFM2, GCH1, NFE2L2, FTH1, FTL, SLC3A2, HSPB1, ACSL3, SCD, DHODH, NQO1, GCLC, GCLM, TXNRD1, SLC40A1, CISD1, FANCD2, AKR1C1

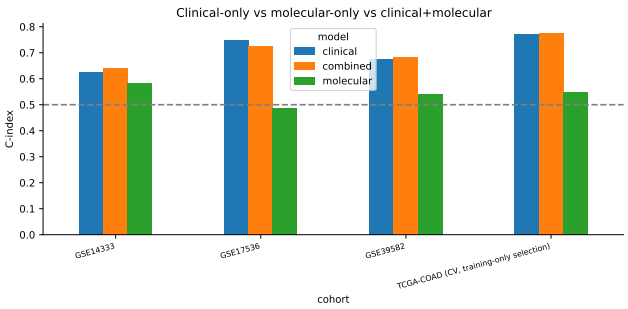


Figure 11: Clinical-only, molecular-only and combined model discrimination by cohort.

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